

Why gradient descent acquires slope scale slowly

Mechanism, empirical evidence, and a population theorem

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The question and the present answer

Our numerical construction obtains accurate approximation using narrow tanh transitions. Training the slopes from small initialization often produces much broader features. **We want to explain why deterministic gradient descent can continue fitting while failing to acquire the scales used by that construction.** The useful conclusion is a rate or a finite-time population bound; it need not assert that every slope shrinks or that training reaches a trapping equilibrium.

The strongest empirical finding is that, after the initial coarse fit, slope motion is driven almost entirely by the **effective fine force**: the force from the remaining errors, including the compensation needed to maintain the constant and linear fit. The departure from this balance is small in the audited GD regime. The remaining question is why the effective force does not rapidly reinforce itself. Wide populations have weak nonlinear sensitivity, and experiments indicate slow geometry feedback even when substantial error remains. Correcting generated lower-order error can further suppress motion, but it is not the sole explanation across targets.

This note derives that reduction, tests its interpretation against the experiments, and proves a conditional population bound. Its substantive open assumption concerns preservation of the population's shape, not an assumed small future force. Existing proofs that derive persistence from initial data are presently too conservative to explain the observed duration. Thus we have a supported mechanism and a precise proof target, not yet a general theorem that GD cannot learn useful slopes on practical horizons.

Notation used throughout. Norms on parameter vectors are Euclidean; function norms use the empirical mean over training inputs.

Symbol	Meaning
a_j, b_j, c_j, d	Signed slope, hidden bias, readout weight, and output bias.
$W, h = 2/N_{\text{ref}}$	Actual neuron count and construction spacing; W includes halo neurons.
$\gamma_j = a_j , \lambda_j = h a_j $ $g = \nabla L, F, R$	Physical and normalized slope scales. Full gradient, effective fine gradient, and coarse tracking correction. The velocity has the negative sign.
e_C, e_H	Constant/linear residual coefficients and their full orthogonal complement.
M, C_6	Total particle second moment and normalized sixth moment, defined in Section 6.

The empirical starting point. Across thirteen targets, the orange tracking force drops below the blue effective fine force after the initial transient. Black then follows blue. The derivation below explains this separation; the later question is why the surviving force often produces so little scale travel.

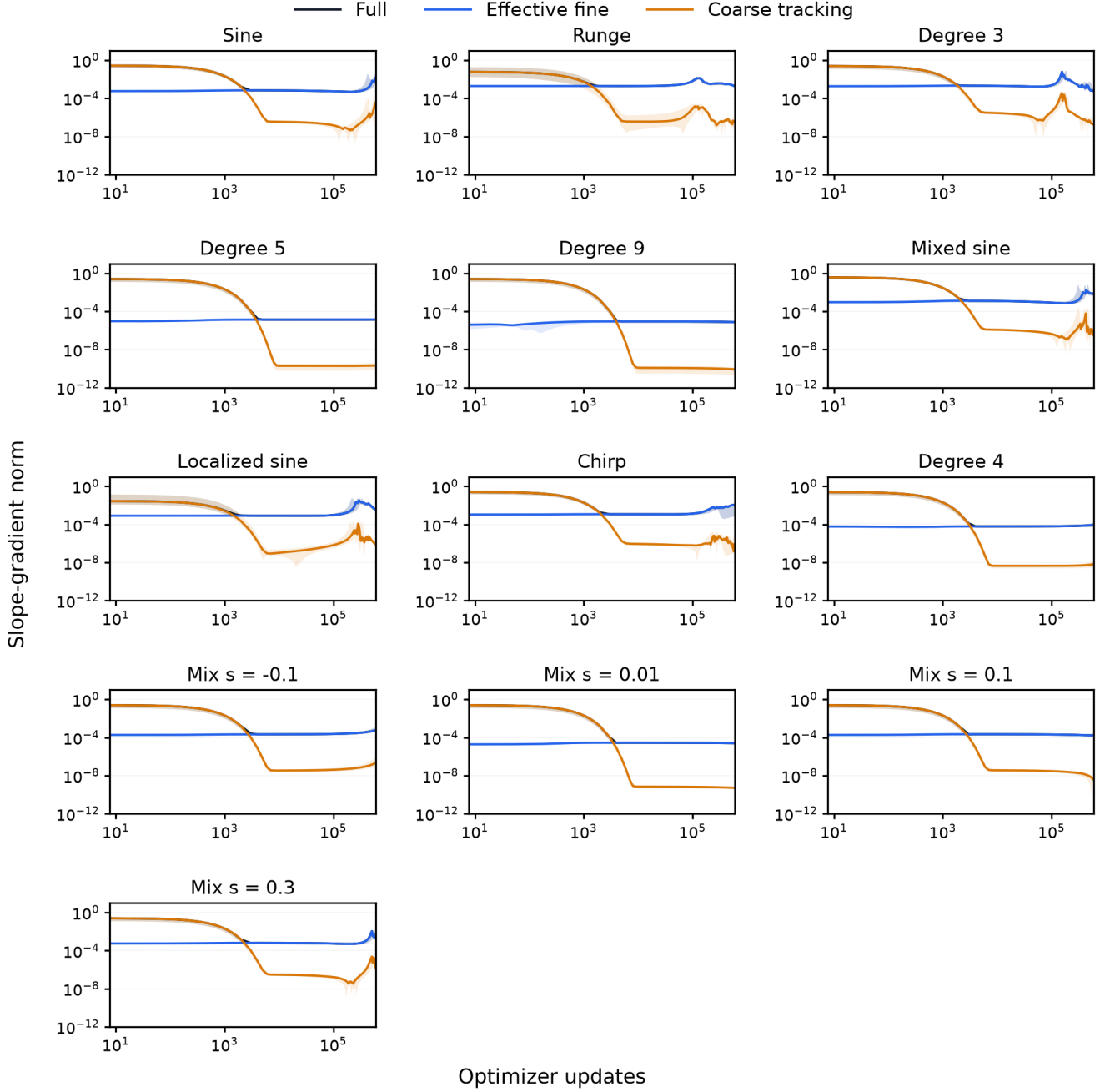


Figure 1: Full slope-gradient norm (black), effective fine-force norm (blue), and coarse-tracking norm (orange) during ordinary GD. Each panel shows five seeds; lines are medians and shading is the full seed range, not a confidence interval. Width 177, 2,048 training points, learning rate 0.002, through 600k updates. The norm axis is logarithmic; the update axis is logarithmic above update 10. Blue includes balanced coarse compensation. The vectors add, but their norms need not. These are sampled force magnitudes, not outward-motion or acquisition measurements. Target labels are defined in Appendix B.1.

1. Start with the coupled training dynamics

Example and objective. At reference resolution $N_{\text{ref}} = 512$, $\lambda = 0.25$ means $|a| = 64$. At $N_{\text{ref}} = 1024$ it means $|a| = 128$. The question therefore concerns scale relative to resolution, rather than a fixed physical slope. The numerical construction places transitions on a grid of spacing h and uses $|a|$ of order $1/h$ to balance resolution and conditioning. The particular threshold 0.25 is an informative construction benchmark, not a proved necessary or sufficient condition for every possible approximation.

For inputs $x_i \in [-1, 1]$, let

$$f_\theta(x) = d + \sum_{j=1}^W c_j \tanh(a_j x + b_j), \quad r = f_\theta - y, \quad L(\theta) = \frac{1}{2} \langle r, r \rangle_m,$$

where $\langle u, v \rangle_m = m^{-1} \sum_i u(x_i) v(x_i)$. Writing $u_j = a_j x + b_j$, differentiation gives

$$\begin{aligned} g_{a,j} &= c_j \langle r, x \operatorname{sech}^2 u_j \rangle_m, & g_{b,j} &= c_j \langle r, \operatorname{sech}^2 u_j \rangle_m, \\ g_{c,j} &= \langle r, \tanh u_j \rangle_m, & g_d &= \langle r, 1 \rangle_m. \end{aligned} \tag{1}$$

The gradient-flow ODE is $\dot{\theta} = -g(\theta)$. Ordinary full-batch GD is $\theta_{n+1} = \theta_n - \eta g(\theta_n)$, with every component evaluated at the same old state. In particular, g_c **always means the readout gradient**; it does not denote a coarse gradient. These are noiseless coupled dynamics: readouts influence slope sensitivity, while moving slopes and biases changes which errors the readouts can fit.

The persistence statements below restart the clock at an observed post-transient checkpoint. They do not attempt to explain the initial coarse-fitting phase or prove entry into the assumed regime.

Equation (1) already explains why a large residual need not produce large slope movement. Its contribution depends on its correlation with the current tangent feature $c_j x \operatorname{sech}^2 u_j$. Better frozen-geometry readout fitting can reduce error without acquiring the intended slope regime. Such improvement is not evidence of precision approximation or of successful geometry learning.

Prediction to test. Identify the error components that actually produce slope motion, then explain how their sensitivities evolve. A loss curve alone cannot distinguish small driving error from poor coupling to a large error.

2. Separate coarse tracking from the force that survives balance

Motivating observation. Figure 1 makes the reduction visible: after the transient, the full slope force follows the effective fine force, including its later recovery on some targets. This is not a claim of universal force decay. Across 60 starts on ten additional target functions, the slope-gradient correction beyond the effective fine force is small after coarse fitting. We need a decomposition that retains what coarse fitting still does to the effective dynamics; simply deleting the constant and linear residual from (1) would miss that effect.

Choose a fixed empirical orthonormal basis with coarse space $C = \operatorname{span}\{1, x\}$. Let e_C be its two residual coefficients, and e_H the full orthogonal complement. We write $\|e_H\|_m$ for the norm of the reconstructed fine residual, equal to its coefficient-vector norm. Write $J_C = D_\theta e_C$ and $J_H = D_\theta e_H$. Transposes below are adjoints in these orthonormal coordinates. Then

$$g = J_C^T e_C + J_H^T e_H.$$

If polynomial basis functions are used, a *quadratic error* is the coefficient along the degree-two orthogonal polynomial, and a *cubic error* is the degree-three coefficient. This is an exact decomposition of residual shapes, not a Taylor approximation. A Taylor approximation of the activation enters separately in Section 5.

Assume $K = J_C J_C^T$ is invertible. Define

$$\begin{aligned} g_H &= J_H^T e_H, & \ell &= K^{-1} J_C g_H, \\ \Pi &= I - J_C^T K^{-1} J_C, & F &= \Pi g_H, \\ z &= e_C + \ell, & R &= J_C^T z. \end{aligned} \tag{2}$$

These definitions give the **exact identity** $g = F + R$ and $J_C F = 0$. The effective force $F = g_H - J_C^T \ell$ includes the compensating coarse response. The variable z measures departure from that instantaneous balance: $\dot{e}_C = -Kz$. Thus small z is a tracking statement. It is neither $e_C = 0$ nor optimal least-squares fitting of every readout.

At exact balance, $R = 0$ but the compensation $-J_C^T \ell$ usually remains. This distinction is essential: **small coarse disequilibrium does not mean coarse compensation is irrelevant**. Also, $z = 0$ is not automatically an invariant manifold of full GD. The reduced flow $\dot{\theta} = -F$ is a model whose discrepancy must be controlled.

The remaining errors evolve according to

$$\dot{e}_H = -S e_H - J_H J_C^T z, \quad S = J_H \Pi J_H^T \succeq 0. \tag{3}$$

Equations (2)-(3) expose the two corrections to audit: direct tracking force on slopes and tracking's effect on the errors driving later motion. They also show why the model remains coupled when tracking is small: S , Π , and J_H all change with the parameters.

Some experiments retain polynomial degrees only through 65. Their identity instead includes an omitted-residual term $g_\perp$; their reported remainder combines tracking and this term. The full-complement formulas used in the theorem and the later population audit have no omitted-residual term.

Prediction and evidence. The ten new functions comprise two each of exponentials, off-center Gaussians, compact bumps, tanh steps, and continuous kinks. Two seeds and forks at updates 100k, 400k, and 600k give 60 starts at $W = 177$, with $m = 2048$ and $\eta = 0.002$. The functions were fixed before their outcomes were inspected.

Table 1. Effective-force dominance on the new-function panel. Each row summarizes the same 60 ordinary-GD continuations. Ratios compare Euclidean slope-gradient norms, except the final row, which compares signed sums.

Diagnostic	Median	Maximum
Remainder / effective slope force at the fork	0.0206%	1.03%
Same ratio after 200k additional updates	0.0103%	0.0958%
Accumulated signed remainder-travel / effective-travel vector norms	—	0.389%

These measurements support force attribution, not uniform bounds between saved states. Signed cancellation also makes the last row different from an absolute travel budget. A broader cross-function audit of 1,665 saved states from 333 starts found direct tracking slope-force ratios at most 1.24%, and tracking-induced fine-residual forcing ratios at most 0.70%. The latter compares the norms of the two right-hand-side terms in (3). Neither study turns 200k into a universal barrier time.

3. What keeps the effective force weak?

Example. At width 1409, removing the model's residual-relaxation feedback changes positive normalized slope travel over 20k additional updates by a median 0.00505%. Doubling geometry feedback changes it by 1.51%. The arms match the entire gradient at the initial fork and retain tracking evaluated at their own states. This comparison concerns how the force changes, rather than giving one arm a larger initial push.

For the pure effective flow, put $T = \Pi J_H^T$, so $F = Te_H$. Differentiating along $\dot{\theta} = -F$ gives

$$\frac{d}{dt} \frac{\|F\|^2}{2} = - \underbrace{\|J_H F\|^2}_{\text{residual relaxation}} - \underbrace{F^T (DT[F]) e_H}_{\text{geometry and compensation feedback}}. \quad (4)$$

The first term always depletes force. The second is signed: changing features, readouts, biases, and coarse compensation can reinforce or suppress it. Full flow adds $-F^T DF[R]$. Equation (4) identifies a mechanism to investigate, rather than imposing a favorable sign on all motion.

There are two compatible routes to slow acquisition. **Generated-error correction** can oppose expansion: a broad feature creates unwanted quadratic or cubic output, and reducing that error can be easier than resolving the target’s finer structure. **Weak reinforcement** can also keep motion small when those errors barely relax: the population’s sensitivity itself changes too slowly to produce a much stronger force. The latter matters for sine and other targets without degree-nine orthogonality.

For a concrete local Taylor illustration of generated-error correction, take one zero-bias feature, hold its readout c fixed, and retain its cubic coefficient $A(a) = -\chi ca^3$, with $\chi > 0$ set by the polynomial normalization. If the target has no cubic component, the cubic-error loss $A(a)^2/2$ alone gives $\dot{a} = -3\chi^2 c^2 a^5$, which shrinks either sign of slope. A nonzero target cubic component changes this force, and joint training adds the coupling in (2). This scalar illustration explains a possible contribution, not the net direction of the full population.

Table 2. Experiments distinguish slow motion from a universal restoring force. These are finite-window, matched comparisons; they do not establish a sign rule for every neuron or target.

Test	Observation	Consequence for the explanation
Feedback interventions, 12 cases at $W = 1409$, 20k updates	Residual-relaxation removal: 0.00505% median travel change; doubled geometry feedback: 1.51%	Geometry feedback matters more here, but neither perturbation produces rapid reinforcement.
Small outward pulses, 23 functions and two seeds	At the smallest tested amplitude, 99.0-100.3% of the directional offset remains after 20k updates	Strong restoration is absent in the tested directions.
Exact fourfold neuron cloning, development cohort	Ordinary cloned motion is about 0.275 of original; compensating geometry mobility gives about 1.05; compensating only readout mobility changes little	A universal readout-dissipation explanation fails this controlled comparison.

The cloning test preserves the represented function by splitting each readout across identical copies. It divides each copy’s geometry mobility by four while multiplying aggregate readout mobility by four. Compensating both recovers the original trajectory to the reported numerical precision. This separates two effects that width changes alone would confound. Readouts still matter through the coupled force; the test rejects their accelerated fitting as a sufficient explanation of the clone slowdown.

The feedback study contains 40 forks and 240 branches. The outward pulses match coarse outputs, coarse disequilibrium, and the slope gradient to first order; symmetric pulses at halved amplitudes check the finite perturbation. Separate physical parameter kicks provide a caution: first-order matching can still create substantial tracking at finite amplitude. Those contaminated kicks are not clean tests of a low-tracking mechanism.

The refined prediction. Wide, diffuse populations should exhibit weak reinforcement even when some slopes expand and the fine residual remains almost fixed. A useful theory should bound collective travel without requiring universal contraction, common signs, or a stable equilibrium.

4. The population is heterogeneous, but its motion is small

Example. In the width-1409 panel, particles violating an earlier theorem’s alignment/readout-dominance condition constitute a median 71.5% of the population and account for 70.2% of outward travel. Yet the largest endpoint

increase in the population maximum of λ is only 9.43×10^{-6} over 20k updates. These are ordinary, slowly moving populations, not an aligned population with a negligible exceptional set.

The population audit contains 223 distinct checkpoints from 23 target instances. None satisfies the old aligned, zero-hidden-bias sector: after choosing neuron orientations, that sufficient regime requires $c_j \geq a_j \geq 0$ and $b_j = 0$ for every neuron. The median negative slope-readout product fraction is 44.6%; hidden biases remain substantial in rescaled coordinates. This falsifies that sector as an explanation of these states, while leaving its conditional mathematical result intact.

There is nevertheless a coherent width pattern. Set $X_j = (\alpha_j, \beta_j, \zeta_j) = \sqrt{W}(a_j, b_j, c_j)$. At three independently initialized widths, typical rescaled slopes stay comparable, whereas physical slope forces shrink strongly with width.

Table 3. Width scaling at the 20k-update forks. Medians over six targets and two seeds at each width; force values use the retained modal audit.

N_{ref}	Actual W	RMS($\sqrt{W}a$)	$W^{3/2}$ RMS(F_a)
128	177	1.456	0.628
512	705	1.471	0.695
1024	1409	1.466	0.661

The six functions are degree five, mixed sine, an off-center Gaussian, a compact bump, a tanh step, and a kink. No case reaches $\lambda = 0.25$ through 40k total updates. Three widths support this regime; they do not establish an asymptotic law. The latest audit also finds that, at $W = 1409$, the endpoint full-force norm after 20k further updates is only 0.984-1.045 times its initial value across the twelve paths. Endpoint comparisons do not bound intervening peaks. They identify the persistence that a theorem must explain.

Mechanistic interpretation. Fine motion depends on mixed population moments, not just slope size or a single readout norm. For example, $\mathbb{E}_W[\alpha\zeta]$ controls a leading linear coefficient, whereas $\mathbb{E}_W[\alpha^3\zeta]$ helps control a cubic coefficient. Positive and negative groups can affect these moments differently. Hidden biases add more mixed moments. Shared coarse compensation couples all groups, so a group carrying outward motion may simultaneously supply compensation opposing motion elsewhere. Discarding all misaligned neurons loses much of the mechanism.

Prediction. Control a few population moments and their correlations. The conclusion should concern the fraction of neuron labels that ever travel far enough, allowing individuals to move in different directions.

5. A useful coupled surrogate, with a visible domain of validity

Example. At supplied wide-network states, a quintic approximation to tanh reconstructs the exact effective slope force with median relative errors of 0.0396% at width 705 and 0.00863% at width 1409. This supports a simpler local description, provided we retain the actual heterogeneous slopes, biases, and readouts.

The approximation is now a Taylor expansion of the activation, $\tanh u \approx u - u^3/3 + 2u^5/15$. With $M_{ijk} = \mathbb{E}_W[\alpha^i \beta^j \zeta^k]$, its output is

$$f_5(x) = d + M_{011} + xM_{101} - \frac{1}{3W} \sum_{r=0}^3 \binom{3}{r} M_{r,3-r,1} x^r + \frac{2}{15W^2} \sum_{r=0}^5 \binom{5}{r} M_{r,5-r,1} x^r. \quad (5)$$

After removing affine output, the leading nonlinear output carries a factor $1/W$. This is the origin of weak fine sensitivity in the diffuse regime. The surrogate evolves its parameters and recomputes its compensation, rather than freezing an initial Jacobian. It is still a coupled dynamical system; a closed system for just a few moments would require additional justification.

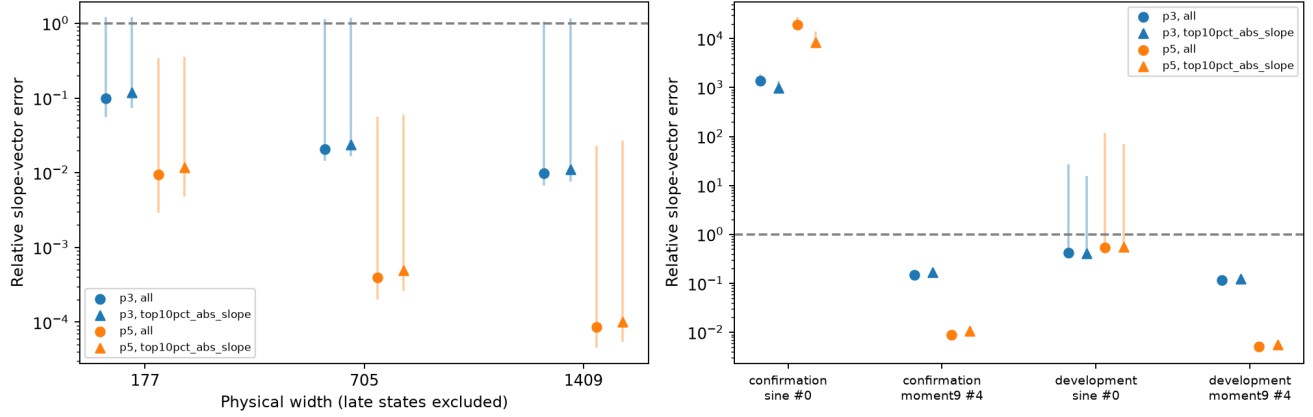


Figure 2: Quintic force accuracy is strong in the wide panel but fails in late sine. Each point is a median and each segment a full range. Left: 96 sampled states per width, comprising twelve cases and eight horizons. Right: four late cases kept separate; source indices in the labels are archive indices, not seeds. Blue is cubic, orange quintic; triangles restrict to the largest 10% of slopes. These are force reconstructions at actual states, not forecasts.

Prediction and separate forecast test. If the evolving surrogate captures the relevant coupling, it should predict motion from its own state without receiving future GD checkpoints. On a separate seed-confirmation panel, the initially anchored quintic field has median relative slope-displacement errors of 0.876%, 0.0547%, and 0.0260% at widths 177, 705, and 1409 over 20k updates. The corresponding constant-force forecast errors are 20.0%, 4.71%, and 2.26%. Anchoring adds a fixed correction computed at the initial state; no future coefficients are fitted. The unanchored quintic errors are 2.86%, 0.0834%, and 0.0324%. These forecasts are stronger evidence than reconstructing force at supplied states, but remain finite-window tests on those functions and seeds.

Late pure sine is an important failure case. Its polynomial force can be wrong by orders of magnitude. Even the exact-force scalar growth forecast can miss because geometry reinforcement declines while residual relaxation strengthens. Thus neither ninth-degree orthogonality nor one frozen growth rate explains all targets. The exact decomposition (2) survives these failures; the polynomial approximation and frozen-feedback models do not have universal validity.

6. A population theorem that exposes the missing assumption

Motivation. We do not need to predict every slope accurately. We need to show that enough slopes cannot travel from their small initial values to λ_* quickly. The width evidence suggests preserving a diffuse population, rather than a common sign pattern. The following proposition makes that idea precise without assuming that the future force is small.

Define the empirical particle moments

$$M = \mathbb{E}_W |X|^2 = \sum_j (a_j^2 + b_j^2 + c_j^2), \quad M_6 = \mathbb{E}_W |X|^6, \quad C_6 = M_6/M^3. \quad (6)$$

C_6 measures concentration: it grows when a small group carries a disproportionate share of the particle magnitude. Uniformly scaling every particle leaves C_6 unchanged. A bound on C_6 therefore does not assume small slopes, a small second moment, or little motion.

Proposition 1: conditional delay under effective flow. Consider $\dot{\theta} = -F$ with the full complement in (2), $|x_i| \leq 1$, and a well-defined coarse projector throughout $[0, T]$. Suppose $M_0 > 0$ and $C_6(t) \leq \bar{C}_6$ on that interval. Let

$$Y_0 = \|e_H(0)\|_m, \quad k = \frac{6Y_0\sqrt{\bar{C}_6}M_0}{W}.$$

For $0 \leq t \leq T$ with $kt < 1$, the total physical parameter path and second moment obey

$$\int_0^t \|F(s)\| ds \leq A(t) := \sqrt{M_0}[(1 - kt)^{-1/2} - 1], \quad M(t) \leq \frac{M_0}{1 - kt}. \quad (7)$$

For $0 \leq \lambda_0 < \lambda_*$, let p_0 be the fraction initially above λ_0 . The fraction of labels ever reaching λ_* by time t satisfies

$$p_{\text{ever}}(t) \leq \min \left\{ 1, p_0 + \frac{h^2 A(t)^2}{W(\lambda_* - \lambda_0)^2} \right\}. \quad (8)$$

Appendix A proves this for exact tanh, not a polynomial substitute. The central step is a structural sensitivity bound,

$$\|F\| \leq \frac{3}{W} \|e_H\|_m \sqrt{M_6}. \quad (9)$$

Affine projection removes the leading Jacobian, leaving a cubic-size remainder; the coarse compensation is an orthogonal projection and cannot increase its full norm. Under effective flow the fine loss decreases. These facts close a differential inequality for M under the shape assumption. Small future force is a consequence, not a premise.

For a desired additional crossing fraction $\varepsilon > 0$, (8) remains at most $p_0 + \varepsilon$ whenever

$$t \leq \frac{1}{k} \left[1 - \left(1 + \frac{\sqrt{\varepsilon W}(\lambda_* - \lambda_0)}{h\sqrt{M_0}} \right)^{-2} \right], \quad (10)$$

within the assumed interval and with $k > 0$. At fixed $kt < 1$, bounded $M_0, Y_0, \overline{C}_6$ give a time scale proportional to W and a small population crossing allowance. If also h is proportional to $1/W$, that allowance in (8) is $O(W^{-3})$ for a fixed normalized gap. This is a conditional width prediction, not an evaluated practical horizon. It is an absolute speed bound; it need not preserve an unusually tiny measured initial force.

Ordinary GD requires its own statement. Suppose the projector (2) exists at each pre-update state of actual GD, and $L_n \leq L_*$, $C_{6,n} \leq \overline{C}_6$, and $\|R_{abc,n}\| \leq \delta_n/W$, with $\delta_n \geq 0$; the subscript excludes the output bias. Set

$$\begin{aligned} \overline{m}_0 &= \sqrt{M_0}, \\ \overline{m}_{n+1} &= \overline{m}_n + \frac{\eta}{W} \left(3\sqrt{2L_*\overline{C}_6} \overline{m}_n^3 + \delta_n \right). \end{aligned} \quad (11)$$

Then $\sqrt{M_n} \leq \overline{m}_n$, and (8) holds through update N with $A(t)$ replaced by $\overline{m}_N - \overline{m}_0$. Appendix A proves this directly from simultaneous GD. Loss descent needs a justified step size, and the tracking assumption concerns all slope, bias, and readout coordinates. Small measured slope tracking alone does not supply that premise. This corollary shows exactly where tracking control enters a population argument; it is not yet a certified bound for the experimental continuations.

Assumptions to earn. Small initial normalized scales and broad particle distributions are observed. Effective-force dominance and slow moment changes are observed in specified windows. But a useful interval-wide bound on C_6 , the required full tracking budget, and their preservation from initial data have not been established by these observations. The proposition names this gap rather than hiding it inside a future-force assumption.

7. What is proved, and what would complete the mechanism?

A concrete comparison. At width 1409 the observed second moment changes by only a factor 0.99790-1.00433 between the endpoints of a 20k-update continuation. Nevertheless, FP64 evaluation of our latest initial-data persistence bounds reaches at most 38 GD updates on the archived panel. That mismatch is a limitation of the sufficient estimates, not a prediction that the dynamics escape after 38 steps.

Table 4. Current proof and evidence status. The different rows answer different questions and should not be combined into one claimed horizon.

Result	What is established	What it does not establish
Exact decomposition and Proposition 1	Effective-force identity; conditional population travel bound with a complete proof	Preservation of the proposed population shape on an informative interval
Generic GD energy baseline	Outward-rounded initial checks for 18 width-panel states exclude $\lambda = 0.25$ for 50k further updates	Why the effective force remains weak; a universal 50k barrier
Existing exact-tanh persistence proof	Derives conditioning, force, and tracking bounds from initial data, allowing biases and mixed signs	Useful constants: only 22/223 states admitted, all at width 1409; best refined interval 38 updates
Coupled quintic forecasts	Accurate 20k motion forecasts on the specified wide confirmation panel	A regional error certificate or a valid late-sine model

The energy baseline is ordinary-GD mathematics, with verified assumptions at the archived 20k states of six targets and three widths. It reaches nominal update 70k and already rules out a distant threshold without any effective-force analysis. Mechanistic theorems must add an explanation of rates, width dependence, or much longer persistence, rather than claiming that threshold exclusion alone validates the mechanism. The latest persistence audit evaluates its bounds in FP64, without interval certification; its short valid windows should be reported alongside the successful observations.

The next theoretical question is preferential growth. For a nonzero particle population, exact differentiation gives

$$\frac{d}{dt} \log C_6 = 6 \left[\frac{\mathbb{E}_W(|X|^4 X \cdot \dot{X})}{M_6} - \frac{\mathbb{E}_W(X \cdot \dot{X})}{M} \right]. \quad (12)$$

Concentration increases when large particles receive disproportionately positive radial motion. Thus the missing shape condition has a concrete mechanistic test: attribute the two weighted radial-growth terms to effective fine correction and shared compensation, retaining mixed signs and biases. An upper bound on their difference, derived from mixed moments of the ODE, would propagate C_6 and close the population argument. This is more useful than controlling every neuron’s position separately. It remains a research target, not a proved invariant or an already measured empirical success.

The discriminating follow-up is to measure (12) on existing trajectories, then perturb concentration or the relevant mixed moments while matching initial coarse balance and force as closely as possible. If stronger preferential growth accelerates acquisition, the shape mechanism gains a causal test. If concentration grows but scale travel remains small, a bound using signed outward coupling or the sensitivity actually reached by the residual should replace the isotropic sixth-moment bound. Either result changes the theorem’s assumptions; it is not a search for smaller constants alone.

Scope across targets and optimizers. The evidence extends beyond degree nine and sine: 23 target instances in the broad audit, ten separately chosen new functions in the force test, and six families in the width/intervention panel. Repeated checkpoints are not independent target samples, and the wide-panel conclusions have not been established for all 23 functions. The horizon is part of each statement. Eventual movement after millions of updates does not refute a finite-time slowdown, and long-run forecast failures do not define its practically relevant duration.

Adam needs a different dynamical theorem. Its tracking activity can be large in raw gradients and component step lengths while largely cancelling in net outward motion. The Adam attribution experiments measure these separately and also contain escapes in some regimes. Momentum and adaptive scaling change both the metric and the state. The Euclidean GD bound above does not transfer to Adam merely because some of its trajectories also show scale saturation.

Appendix A. Proofs and the transport interpretation

A.1. Exact-tanh sensitivity after removing affine output

Let P_H be the orthogonal projection, in the empirical function norm, onto the complement of $\text{span}\{1, x\}$. The output Jacobian columns for one neuron are $cx \text{sech}^2 u$, $c \text{sech}^2 u$, and $\tanh u$. The corresponding affine columns are cx , c , and u ; P_H annihilates them, as it does the output-bias column 1.

For every real u ,

$$|1 - \text{sech}^2 u| = \tanh^2 u \leq u^2, \quad |\tanh u - u| \leq |u|^3/3.$$

The second inequality follows by integrating $\tanh^2 s \leq s^2$ from zero to u . With $s_j^2 = a_j^2 + b_j^2 + c_j^2$ and $|x| \leq 1$, $|u_j|^2 \leq 2s_j^2$. Summing squared column errors gives

$$c_j^2(1 + x^2)u_j^4 + u_j^6/9 \leq 8s_j^6 + \frac{8}{9}s_j^6 = \frac{80}{9}s_j^6.$$

Projection is contractive, so the Hilbert-Schmidt bound, hence also the operator bound, is

$$\|J_H\|^2 \leq \frac{80}{9} \sum_j s_j^6 = \frac{80}{9W^2} M_6 \leq \frac{9}{W^2} M_6. \quad (\text{A1})$$

Because Π is an orthogonal parameter-space projector, $\|F\| = \|\Pi J_H^T e_H\| \leq \|J_H\| \|e_H\|_m$, proving (9). No parity, sign alignment, or individual support cap is required. The inequality is global for $\tanh$, although useful scaling requires controlled moments.

A.2. Closing the effective-flow moment bound

Under $\dot{\theta} = -F$,

$$\frac{d}{dt} \frac{1}{2} \|e_H\|_m^2 = -e_H^T J_H \Pi J_H^T e_H = -\|F\|^2 \leq 0. \quad (\text{A2})$$

Let $p = (a, b, c)$, so $M = \|p\|^2$. Using (9) and the shape assumption,

$$\dot{M} = -2p^T F_{abc} \leq 2\sqrt{M} \|F\| \leq \frac{6Y_0 \sqrt{C_6}}{W} M^2. \quad (\text{A3})$$

Comparison with the scalar equality gives $M(t) \leq M_0/(1 - kt)$. Substituting this upper bound in $\|F\| \leq 3Y_0 \sqrt{C_6} M^{3/2}/W$ and integrating gives (7). If $Y_0 = 0$, then $F = 0$ and the trajectory is stationary; (7) is read with $A = 0$. The proposition assumes the coarse solve remains defined and does not derive that premise from (A3).

A.3. From collective travel to ever-acquired labels

For each neuron set $v_j(t) = \int_0^t |\dot{a}_j(s)| ds$. Minkowski's inequality gives

$$\left(\sum_j v_j(t)^2 \right)^{1/2} \leq \int_0^t \|\dot{a}(s)\| ds \leq A(t).$$

A label initially satisfying $h|a_j(0)| \leq \lambda_0$ can reach λ_* only if $v_j(t) \geq (\lambda_* - \lambda_0)/h$. Counting these labels by their squared travel gives (8), including crossings followed by a return below threshold. It is therefore stronger than an endpoint count. Solving $h^2 A^2 / [W(\lambda_* - \lambda_0)^2] \leq \varepsilon$ gives (10). No assumption on the direction of individual velocities was used.

A.4. Direct discrete-GD proof

At each iterate the complete empirical residual obeys $\|e_{H,n}\|_m \leq \sqrt{2L_*}$. The exact identity and (A1) imply

$$\begin{aligned} \|p_{n+1} - p_n\| &\leq \eta(\|F_{abc,n}\| + \|R_{abc,n}\|) \\ &\leq \frac{\eta}{W} \left(3\sqrt{2L_*C_6} \|p_n\|^3 + \delta_n \right). \end{aligned}$$

The right side is monotone in $\|p_n\|$. The triangle inequality and induction give $\|p_n\| \leq \bar{m}_n$. Summing the increment bounds then gives $\sum_{n \leq N} \|p_{n+1} - p_n\| \leq \bar{m}_N - \bar{m}_0$. The discrete version of A.3 proves the asserted population count. This proof uses the actual GD step; it does not replace t by ηN in a continuous-time theorem. Loss descent, full tracking control, and shape persistence remain explicit premises to verify or derive separately.

A.5. What the transport PDE contributes

Associate each neuron with $\xi_j = (a_j, b_j, c_j)$ and let $\rho_t = W^{-1} \sum_j \delta_{\xi_j(t)}$. For gradient flow, its weak transport equation is

$$\partial_t \rho + \nabla_\xi \cdot (\rho v[\rho, d]) = 0, \quad f(x) = d + W \int c \tanh(ax + b) d\rho, \quad \dot{d} = -\langle r, 1 \rangle_m. \quad (\text{A4})$$

Here v consists of the three negative gradients in (1). It depends on ρ through the shared residual. Effective flow replaces both particle velocities and the output-bias equation by the corresponding components of $-F$, incorporating the shared coarse projector. The PDE is nonlinear, with no diffusion because training is deterministic. For the finite empirical measure it is an exact rewriting of the particle ODEs, not an additional limiting assumption.

Its useful contribution is to organize the theorem around transported mass, moments, and accumulated action. Equation (A3) bounds a population moment; (8) bounds mass whose characteristics ever cross a scale threshold. Transport alone does not prevent concentration or reinforcement. The missing step is a property of this velocity field, captured by (12), rather than a generic consequence of noiseless transport. Discrete GD transports the empirical measure by a map at each update; A.4 supplies the corresponding population bound without introducing artificial diffusion.

Appendix B. Experimental methods and coverage

B.1. Training setup and target functions

The width studies use 2,048 equally weighted midpoint inputs $x_i = -1 + 2(i + 1/2)/2048$, $i = 0, \dots, 2047$, and binary64 arithmetic. All parameters are trained by simultaneous, deterministic full-batch GD with $\eta = 0.002$. Initially, a, b, c are independent uniform draws from $[-\sqrt{6/(W+1)}, \sqrt{6/(W+1)}]$, with $d = 0$. Initializations are paired across targets at a fixed width and seed, and generated independently across widths. We use $(N_{\text{ref}}, W) = (128, 177), (512, 705), (1024, 1409)$. The difference includes the extra neurons used beyond the reference grid.

Table B1. Six targets in the width and feedback panels. Except for the already normalized polynomial target, each listed function is divided by its RMS on the original training grid. That normalization stays fixed during training and evaluation; target means are not subtracted for training.

Target	Function before RMS normalization
Degree five	$0.3q_0 + 0.4q_1 + \sqrt{0.75}q_5$; q_k are empirical orthonormal polynomials with positive leading coefficient
Mixed sine	$\sin(2\pi x) + 0.5 \sin(6\pi x) + 0.25 \sin(14\pi x)$
Off-center Gaussian	$\exp(-((x + 0.35)/0.22)^2)$
Right compact bump	$\exp(1 - 1/(1 - u^2))$ for $ u < 1$, otherwise zero; $u = (x - 0.35)/0.22$
Right step	$\tanh(14(x - 0.31))$
Kink	$ x + 0.23 $

The degree-five label thus denotes a prescribed orthogonal residual shape, not the monomial x^5 . The same distinction applies to the degree-nine control. The development width panel has two seeds and starts its continuation after 20k updates; its 20k further updates end at 40k total. The quintic confirmation repeats the six targets and three widths with two new seeds. It checks seed variation, not generalization to unseen functions. Where independent-grid fitting errors are measured, evaluation uses 8,192 midpoint inputs; the force and trajectory quantities here concern the training grid.

The thirteen targets in Figure 1. Sine is $\sin(2\pi x)$, and Runge is $1/(1 + 25x^2)$; these two functions are not RMS-rescaled. Degree d , for $d \in \{3, 4, 5, 9\}$, denotes $0.3q_0 + 0.4q_1 + \sqrt{0.75}q_d$, using the same empirical orthonormal polynomials as Table B1. Each of these polynomial targets has unit empirical RMS.

Let $S(x) = \sin(2\pi x) + 0.5\sin(6\pi x) + 0.25\sin(14\pi x)$. Mixed sine is $S(x)$ divided by its training-grid RMS. Localized sine is $S(x)\exp[-(x/0.4)^2/2]$, divided by its own RMS. Chirp is $\sin[2\pi(u + 4u^2)]$, with $u = (x + 1)/2$, likewise divided by its own RMS. All these normalizers are fixed from the original 2,048 training points. Finally, the four panels labeled Mix s use

$$y_s(x) = 0.3q_0(x) + 0.4q_1(x) + \sqrt{0.75} [sq_3(x) + \sqrt{1-s^2}q_9(x)], \quad s \in \{-0.1, 0.01, 0.1, 0.3\}.$$

These blends already have unit empirical RMS. Figure 1 uses five seeds numbered 0 through 4, with the same width-177 initialization law and training grid described above. Its full-complement decomposition has no omitted modal-residual term. At each saved state, the median and range are taken across the five seeds separately for each force norm. The shading is neither a confidence interval nor a bound on the unsampled times between states.

B.2. What the interventions change

The feedback experiment separates two evolving contributions while preserving the initial force. At a fork θ_0 , save $e_{H,0}$ and F_0 . At the current state define

$$A(\theta) = \Pi(\theta)J_H(\theta)^T e_{H,0}, \quad B(\theta) = F(\theta) - A(\theta), \quad C(\theta) = \Pi(\theta)F_0.$$

Use the modified gradient

$$g_{\kappa,\nu} = g + (\kappa - 1)(A - C) + (\nu - 1)B. \tag{B1}$$

At the fork, $A = C = F_0$ and $B = 0$, so all arms have the same gradient. The six choices $(\kappa, \nu) = (1, 1), (0, 0), (0, 1), (2, 1), (1, 0), (1, 2)$ respectively give ordinary GD, a projected constant field, removed or doubled geometry feedback, and removed or doubled residual-relaxation feedback. Each arm retains its own current tracking correction. These are controlled vector-field interventions, not all gradients of a common modified loss.

For the pulse experiment, an outward parameter direction lies in the common nullspace of the derivatives of coarse output, coarse disequilibrium, and the actual slope gradient. Symmetric amplitudes $\pm 0.01, \pm 0.005, \pm 0.0025$ are measured in the experiment's relative block-RMS units. The offset-retention numbers in Table 2 use the smallest amplitude. Halving checks distinguish the predicted first-order response from finite-amplitude matching errors.

For cloning, k identical copies each receive readout c_j/k . The function and residual are initially unchanged. Replica symmetry makes the geometry rate $1/k$ of the original and the aggregate readout rate k times the original. Multiplying geometry learning rates by k , or dividing readout learning rates by k , isolates these effects. The construction spacing h is held fixed because this test changes parameterization, not resolution.

B.3. Coverage and numerical meaning

Table B2. Evidence roles. These studies overlap. Their counts must not be added and interpreted as independent samples of target functions.

Study	Design and role
Opening force figure	Thirteen targets and five seeds; 65 ordinary-GD trajectories through 600k updates at width 177. Full-complement decomposition at saved states.
New-function force test	Ten additional functions in five families, two seeds, three forks; 60 starts. Functions fixed before inspection of their outcomes.
Width development / confirmation	Six targets, two seeds, three widths per panel; 36 cases each. Starts at 20k, followed for 20k further updates.
Feedback interventions	36 width cases plus four late sine/degree-nine starts; six arms each, 240 branches.
Pulse and cloning tests	23 target instances in separate development and confirmation cohorts. Matched-fork comparisons.
Population audit	223 deduplicated static states; 40 natural continuations with eight snapshots each. Retrospective structural and approximation tests.
Persistence-bound audit	The same 223 states; seven fixed region choices per state. Duration selected from initial information.
Energy verification	Eighteen initial states: six targets, three widths, one seed. Outward rounding checks sufficient GD assumptions.

The ordinary floating-point diagnostics support the reported empirical comparisons, not interval-wide rigorous bounds. In particular, sampled force ratios do not control their maxima between snapshots. The energy verification is different: outward-rounded checks establish its sufficient initial inequalities for exact-real GD starting from the stored binary64 data. It does not certify all earlier training or every rounding error in later machine updates.

For completeness, the generic energy argument underlying Table 4 uses a descent factor $\mu > 0$ satisfying $L_{n+1} \leq L_n - \eta\mu\|g_n\|^2$. Summation and Cauchy-Schwarz give

$$\sum_{n < N} \|\theta_{n+1} - \theta_n\| \leq \sqrt{\frac{\eta N L_0}{\mu}}. \quad (\text{B2})$$

Bounding the loss Hessian along outgoing steps closes a bootstrap ensuring descent. For the eighteen stated checkpoints, outward-rounded initial bounds give $\|(a, b, c)\| \leq 3$, $L_0 < 0.417$, and $\eta = 0.002 < 1/490$. Here is the needed curvature control. On a radius-eight travel region, $\|(a, b, c)\| \leq 11$ and the output Jacobian has norm below 16. The next step has length below $16\sqrt{2}/490 < 0.05$, so it lies in a padded radius-8.05 region. There the squared Jacobian norm is below 246, the output Hessian norm below 19, and the residual norm along the outgoing step below 2.18. These estimates follow from $|\tanh u| \leq |u|$, $|\tanh' u| \leq 1$, $\sup |\tanh''| = 4/(3\sqrt{3})$, and $\|Df\|^2 \leq 1 + 2\|(a, b, c)\|^2$. Consequently $\|D^2L\| < 246 + 19(2.18) < 288$, giving $\mu \geq 1 - 144/490 > 0.7$. The resulting travel bound is $\sqrt{\eta(50,000)(0.417)/0.7} < 8$; all slopes therefore remain below $3 + 8 = 11 < 16$. Since $N_{\text{ref}} \geq 128$, this excludes $\lambda = 0.25$. It illustrates how a distant-threshold statement can hold without identifying the fine-force mechanism.

The scientific question left for review is whether a condition such as (12) can be derived and shown to persist on the observed mixed-sign populations. Equations (7)-(11) state what that would buy: a rate and an ever-acquired population bound, without an accurate forecast of every neuron.